



SERANGOON JUNIOR COLLEGE

2013 JC2 PRELIMINARY EXAMINATIONS

MATHEMATICS

Higher 1

8864

Wednesday

21 August 2013

Additional materials: Writing paper

List of Formulae (MF15)

TIME : 3 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on the cover page and on all the work you hand in.

Write in dark or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

Total marks for this paper is 95 marks

This question paper consists of 6 printed pages (inclusive of this page) and 2 blank pages.

Section A: Pure Mathematics [35 marks]

- 1 Find algebraically the set of values of k for which $(k-15)x^2 - 16x + k - 3 < 0$ for all real values of x . [4]

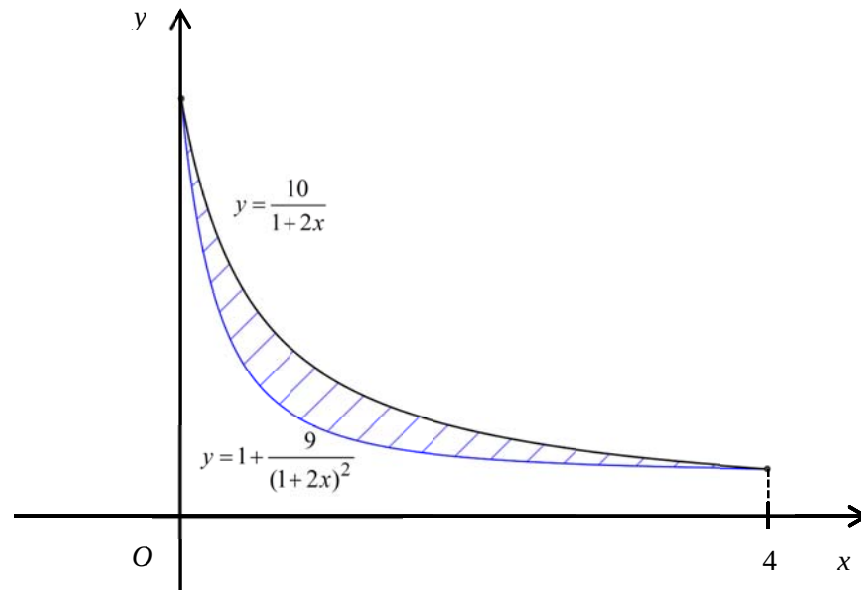
- 2 Solve algebraically the simultaneous equations

$$\frac{1}{2}(2^y) = 4^{x+2},$$

$$\frac{3^y}{3^{xy}} = 27.$$

[4]

3



The diagram shows a sketch of part of the curve $y = \frac{10}{1+2x}$ and part of the curve $y = 1 + \frac{9}{(1+2x)^2}$, with their points of intersection at $x = 0$ and $x = 4$.

Use integration to find the exact area of the shaded region. [4]

- 4 Sketch on a single diagram, the graphs of $y = \frac{x-1}{2-x^2}$ and $y = e^{x-1} - 1$, indicating clearly the equations of any asymptotes for each of the graphs. [3]

Hence find the range of values of x such that $\frac{x-1}{2-x^2} \geq e^{x-1} - 1$. [2]

- 5 (a) Find
- (i) $\int e^{3-2x} dx$ [1]
- (ii) $\int \left(\frac{x^2 - 2x}{\sqrt{x}} \right) dx$. [2]
- (b) Show that $\frac{d}{dx} (e^{-x+\ln x}) = (1-x)e^{-x}$.
- Hence show that $\int_1^4 (x-1)e^{-x} dx = \frac{1}{e^4} (e^3 - 4)$. [5]
- 6 The equation of a curve is $y = \ln \left(\frac{20x}{10-x} \right)$, where $0 < x < 10$.
- (i) Show that $\frac{dy}{dx} = \frac{10}{x(10-x)}$. [2]
- (ii) The gradient of the curve at the point P is $\frac{2}{5}$. Show that the y – coordinate of P is $\ln 20$ and hence find the exact equation of the tangent to the curve at P . [4]
- (iii) The tangent to the curve at P cuts the x – axis at Q . Find the exact coordinates of Q . [1]
- (iv) R is the point on the x -axis vertically below P . Show that the area of triangle PQR is $\frac{5}{4}(\ln 20)^2$. [3]

Section B: Statistics [60 marks]

- 7 (a) A bag contains 3 red balls, 4 green balls and 5 blue balls. Three balls are drawn from the bag at random without replacement. Calculate the probability that
- (i) all three balls are red; [1]
- (ii) the third ball is red; [2]
- (iii) the first two balls are of the same color given that the third ball is red. [3]
- Are the events given in part (i) and part (ii) mutually exclusive? Explain your answer. [1]
- (b) The independent events A and B are such $P(A) = 0.6$ and $P(A' \cap B') = 0.3$. Show that $P(B)$ is $\frac{1}{4}$. [3]

[Turn over

- 8** A promotion at a restaurant entitles diners to a lucky draw for every meal purchased. Based on a survey of 500 randomly chosen diners last year, 411 did not win anything while the rest received a prize.
- (a)** Nine students plan to have their dinner at the restaurant every weekday for one week this year.
 - (i)** Show that the probability that a randomly chosen student will first win a prize on his third meal at the restaurant is 0.120. State an assumption that you have used in your calculation. [2]
 - (ii)** Find the probability that less than three of the students will each first win a prize on their third meal at the restaurant. [2]
 - (b)** Using a suitable approximation, find the probability that at least 100 but less than 190 of the first 1000 diners this year will win a prize. [4]
 - (c)** A disgruntled diner decides to conduct a survey of n diners this year. Find the minimum value of n if the probability that at most 90% of the diners did not win a prize exceeds 0.75. [3]
- 9** A recent survey by a student researcher, of 60 local companies, showed that four in five companies gave retired workers they re-employ the same pay as before. The records at the Labour Ministry, which tracks the practices of all companies in the country, revealed the actual figure to be 60% of all local companies instead.
- (i)** Give a reason for the difference in the student researcher's result if he had used systematic sampling in his selection of companies for the survey. [1]
- A large company with 180 re-employed workers agreed to help the student researcher in his research. He selected 60 re-employed workers from the company for a survey on the work that they do after re-employment. The satisfaction index, x , of each of the worker was recorded.
- It was found that $\sum (x - 5) = 45.1$ and $\sum (x - 5)^2 = 1516$.
- (ii)** Explain how the student researcher could have selected the workers for the survey using the systematic sampling method. [2]
 - (iii)** Find the probability that the mean satisfaction index of a sample of 60 randomly chosen workers lies between 5 and 7.5. [4]
 - (iv)** State two assumptions or approximations used in your calculations in part **(iii)**. [2]

- 10** The mass, x kg, of the contents of each packet in a random sample of 60 cereal packets is measured, and the results are summarized by

$$\sum x = 59.6475, \sum (x - \bar{x})^2 = 15.9772.$$

- (i) Find the unbiased estimates for the population mean and variance. [2]
- (ii) Test, at the 5% significance level, whether the population mean mass of the contents is less than 1.10 kg. [3]
- (iii) Explain the meaning of “at the 5% significance level” in the context of the question. [1]

In another test, a new sample of 30 cereal packets yielded a sample mean of $\bar{x}$ kg. The results are to be used to test, at the 5% significance level, the hypothesis that the population mean mass of the contents differs from 1.10 kg. It may be assumed that the masses of the contents are normally distributed with standard deviation 0.53 kg. Find the range of values of $\bar{x}$ for which the null hypothesis would not be rejected. [4]

- 11** A supermarket sells apples and bunches of bananas. The weight of an apple is known to be normally distributed with mean 150 grams and standard deviation 10 grams. The weight of a bunch of bananas is known to be normally distributed with mean 780 grams and standard deviation 70 grams.

- (i) Find the probability that the total weight of ten randomly chosen apples will exceed twice the weight of a bunch of bananas. [3]

The supermarket sells the apples for \$0.46 each. Each bunch of bananas is sold by weight at a price of \$1.80 per kilogram.

- (ii) Find the probability that two randomly chosen bunches of bananas will cost more than six apples. [4]
- (iii) The supermarket now wants to grade its apples into Grade A, Grade B and Grade C by weight, with Grade A apples being the lightest and Grade C apples being the heaviest.

Find the range of the weight of a Grade B apple if the supermarket wants 15% of its apples to be of Grade A and 20% of its apples to be of Grade B. [3]

- 12** The water temperature T , in $^{\circ}\text{C}$, and the depth D , in metres, were recorded at noon on a certain day at each of 8 locations in a lake. The results are summarized in the table below.

D	10	20	40	80	150	200	250	300
T	25.0	23.5	18.9	14.3	10.2	5.6	4.1	4.0

- (i) Give a sketch of the scatter diagram for the data, as shown on your calculator. [2]
- (ii) Find the product moment correlation coefficient and comment on its value in the context of this question. [2]
- (iii) Find the equation of the regression line of T on D . Sketch this line on your scatter diagram. [2]
- (iv) Calculate an estimate of the water temperature at noon at a place in the region with depth 120 metres. Comment on the reliability of this estimate. [2]
- (v) It has been found that due to equipment error, all readings of temperature are 0.5°C higher than the actual water temperature. Without performing any further calculations, state with a reason, if you would expect a change in the value found in part (ii). [2]

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